PHYSICS GROUP PROJECT REPORT

An Explanatory Guide and Step-by-Step Derivation of: "The Intermediate Axis Theorem (Tennis Racquet Effect)"

A Numerical Study of Torque-Free Euler Rotation for an Asymmetric Rigid Body

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Abstract

This project provides a comprehensive, pedagogical breakdown of the **intermediate axis theorem**—the surprising fact that a freely rotating asymmetric rigid body is *unstable* when spun about its intermediate principal axis, undergoing the periodic 180° flips popularly known as the *tennis racquet effect* or *Dzhanibekov effect*. We follow the empirical-and-numerical treatment of Bubbar & Zhu (2025) [1] and the analytical exposition of Peterson & Schwalm (*American Journal of Physics*, 2021) [2], filling in the intermediate algebraic steps that these and standard texts often compress. Starting from the torque-free Euler equations, we derive the energy and angular-momentum invariants, carry out an explicit linear stability analysis about each of the three principal axes, and show *quantitatively* why only the intermediate axis is unstable. We then verify the theory with a custom fourth-order Runge–Kutta (RK4) integrator coupled to a quaternion orientation solver: the simulation reproduces the flip, conserves energy and $|\mathbf{L}|$ to near machine precision ($\sim 10^{-13}$ relative drift), traces the sphere–ellipsoid *polhode* geometry, and recovers the inverse period–speed scaling law $T = a/(\omega_0 + b) + c$ reported in [1].

Contents

1 Introduction & Contextual Background

The intermediate axis theorem is one of the most counter-intuitive results in elementary rigid-body mechanics. Take a book, a smartphone, or a tennis racquet and toss it so that it spins about the axis with the *intermediate* moment of inertia (for a book, the axis parallel to the text). Instead of spinning cleanly, the object executes an abrupt 180° flip, repeating the flip periodically. Spinning the same object about its longest or shortest axis produces a stable, wobble-free rotation. Remarkably, the motion is entirely *torque-free*: gravity acts at the centre of mass and exerts no torque, so the flip is driven purely by the internal nonlinear coupling of the rotational equations of motion.

1.1 Historical and Pedagogical Significance

The governing equations were written down by Euler in the 18th century, and the geometric (conservation-law) picture dates to Poinsot (1834). Yet the effect remains famously hard to explain intuitively: Richard Feynman puzzled over the "wobbling plate," and the phenomenon entered popular awareness through the 1985 footage of cosmonaut Vladimir Dzhanibekov watching a wing-nut flip in orbit aboard a space station [7]. The topic recurs in the *American Journal of Physics* (AJP) precisely because it is an ideal teaching vehicle: it links abstract tensor algebra to a result one can literally hold in one's hand. Peterson & Schwalm [2] give the exact analytical solution in terms of Jacobi elliptic functions, while Parker & Hunt [6] and Bubbar & Zhu [1] develop the "sphere-and-ellipsoid" conservation argument aimed at introductory students. This report targets the **upper-introductory undergraduate**: we assume familiarity with Newtonian mechanics and basic linear algebra, and we supply every intermediate step that the source papers state as "it can be shown that."

The pedagogical contrast is between two routes to the same conclusion:

- **Core thesis.** For an asymmetric top ($I_1 > I_2 > I_3$), rotation about the largest (I_1) and smallest (I_3) axes is stable, but rotation about the intermediate (I_2) axis is unstable.
- **Traditional textbook route.** Linearise the Euler equations about each axis and inspect the sign of the resulting characteristic exponent (Section 2).
- **The geometric route of the source papers.** Intersect the angular-momentum sphere with the energy ellipsoid in $\mathbf{L}$-space; the intermediate axis sits at a saddle point whose separatrix wraps the entire sphere (Section 4).

1.2 Physical Setup and System Parameters

We model a single rigid body in free space (no external torque) and work in the *body frame* aligned with the three principal axes $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$. In this frame the inertia tensor is diagonal, $\mathbf{I} = \mathrm{diag}(I_1, I_2, I_3)$, and the state of the body is given by its angular-velocity vector $\boldsymbol{\omega}$ together with an orientation quaternion q. Table 1 lists the symbols used throughout.

2 Step-by-Step Mathematical Derivations

This is the core of the project. We take the equations the source papers state compactly and expand every intermediate step.

Table 1: Key symbols, constants, and variables used in the analysis.

Symbol	**Physical Meaning**	**SI Units**
I_1, I_2, I_3	Principal moments of inertia ($I_1 > I_2 > I_3$)	$\mathrm{kg\,m^2}$
$\boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)$	Body-frame angular velocity	rad/s
$\mathbf{L} = \mathbf{I}\boldsymbol{\omega}$	Angular momentum	$\mathrm{kg\,m^2/s}$
E	Rotational kinetic energy	J
$q = [q_0, q_1, q_2, q_3]$	Orientation quaternion (body $\to$ world)	—
Ω	Initial spin rate about the chosen axis (ω_0)	rad/s
ε	Small initial perturbation onto the other axes	rad/s
σ	Linear instability growth rate	$\mathrm{s^{-1}}$
T	Flip / precession period	s

2.1 The Starting Framework: Torque-Free Euler Equations

In an inertial frame, $d\mathbf{L}/dt = \boldsymbol{\tau}_{\text{ext}}$. With no external torque, $\mathbf{L}$ is conserved in the inertial frame. However, the principal axes are fixed to the *rotating* body, so transforming the time derivative into the body frame introduces the kinematic term $\boldsymbol{\omega} \times \mathbf{L}$:

$$\left(\frac{d\mathbf{L}}{dt}\right)_{\text{inertial}} = \left(\frac{d\mathbf{L}}{dt}\right)_{\text{body}} + \boldsymbol{\omega} \times \mathbf{L} = 0. \tag{1}$$

Since $\mathbf{L} = (I_1\omega_1, I_2\omega_2, I_3\omega_3)$ in the principal basis, writing Eq. (1) component by component gives the **torque-free Euler equations**:

$$\boxed{I_1\dot{\omega}_1 = (I_2 - I_3)\,\omega_2\omega_3, \quad I_2\dot{\omega}_2 = (I_3 - I_1)\,\omega_3\omega_1, \quad I_3\dot{\omega}_3 = (I_1 - I_2)\,\omega_1\omega_2.} \tag{2}$$

These are three coupled, *nonlinear* (quadratic) ordinary differential equations. The nonlinearity—each derivative depends on the product of the other two components—is exactly what makes the behaviour non-intuitive.

2.2 The Two Conserved Quantities

Free rotation conserves both kinetic energy and the magnitude of angular momentum. The rotational kinetic energy is

$$E = \tfrac{1}{2}\left(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2\right) = \text{const}, \tag{3}$$

and, writing $L_r = I_r\omega_r$, the squared angular-momentum magnitude is

$$|\mathbf{L}|^2 = L_1^2 + L_2^2 + L_3^2 = I_1^2\omega_1^2 + I_2^2\omega_2^2 + I_3^2\omega_3^2 = \text{const}. \tag{4}$$

Re-expressed in $\mathbf{L}$-space, Eq. (4) is a *sphere* of radius $|\mathbf{L}|$, while Eq. (3) becomes an *ellipsoid*, $L_1^2/I_1 + L_2^2/I_2 + L_3^2/I_3 = 2E$. The motion is confined to the intersection of these two surfaces—the geometric heart of the theorem, revisited numerically in Section 4.

2.3 Linear Stability Analysis About Each Axis

We now answer the central question quantitatively: spin the body *mostly* about one principal axis and ask whether a tiny perturbation grows or merely oscillates.

Derivation Details & Interpretation: Why Only the Intermediate Axis Is Unstable

Case A — spin about the intermediate axis ($\mathbf{r}_2$). Let $\boldsymbol{\omega} = (\eta_1,\ \Omega,\ \eta_3)$ with Ω the large steady spin and $\eta_1, \eta_3 \ll \Omega$ small perturbations. To first order, the middle Euler equation gives $\dot{\omega}_2 \propto \eta_1\eta_3 \approx 0$, so $\omega_2 \approx \Omega$ is constant. The other two equations linearise to

$$\dot{\eta}_1 = \frac{(I_2 - I_3)\Omega}{I_1}\,\eta_3, \qquad \dot{\eta}_3 = \frac{(I_1 - I_2)\Omega}{I_3}\,\eta_1.$$

Differentiating the first and substituting the second eliminates η_3:

$$\ddot{\eta}_1 = \frac{(I_2 - I_3)\Omega}{I_1}\,\dot{\eta}_3 = \underbrace{\frac{(I_1 - I_2)(I_2 - I_3)}{I_1 I_3}\,\Omega^2}_{\equiv\,\sigma^2}\,\eta_1.$$

Because $I_1 > I_2 > I_3$, both factors $(I_1 - I_2)$ and $(I_2 - I_3)$ are *positive*, so the coefficient $\sigma^2 > 0$. The solution is therefore *exponential*, $\eta_1 \sim e^{\pm\sigma t}$, with growth rate

$$\sigma = \Omega\sqrt{\frac{(I_1 - I_2)(I_2 - I_3)}{I_1 I_3}} \ > \ 0 \quad \Longrightarrow \quad \textbf{UNSTABLE}. \tag{5}$$

Case B — spin about the major ($\mathbf{r}_1$) or minor ($\mathbf{r}_3$) axis. Repeating the identical procedure for a steady spin about $\mathbf{r}_1$ gives $\ddot{\eta}_2 = -k\,\eta_2$ with

$$k = \frac{(I_1 - I_2)(I_1 - I_3)}{I_2 I_3}\,\Omega^2 > 0,$$

since both $(I_1 - I_2)$ and $(I_1 - I_3)$ are positive. The right-hand sign is now *negative*, so the solution is *oscillatory*, $\eta_2 \sim \cos(\sqrt{k}\,t)$: the perturbation stays bounded (stable precession). The same is true about $\mathbf{r}_3$, where both differences are negative and their product is again positive. Only the intermediate axis flips the sign of the coefficient and produces real exponential growth.

Equation (5) is the quantitative statement of the theorem: the instability is strongest when the moments are well separated and the spin is fast, and it vanishes ($\sigma \to 0$) as $I_2 \to I_1$ or $I_2 \to I_3$.

2.4 Modelling the Flip Period

The exact period of the fully nonlinear flip is given by Jacobi elliptic functions [2], but Bubbar & Zhu [1] showed that, for initial conditions close to pure $\mathbf{r}_2$ rotation, the precession period T obeys a simple three-parameter *inverse* relation in the initial spin speed:

$$T(\omega_0) = \frac{a}{\omega_0 + b} + c. \tag{6}$$

The leading $1/\omega_0$ behaviour is consistent with Eq. (5), where the instability timescale $1/\sigma$ scales as $1/\Omega$. We test Eq. (6) directly against simulation in Section 4.

3 Physical Approximations and Limiting Cases

Examining limits builds intuition and exposes where the model breaks down.

3.1 The Symmetric-Top Limit ($I_2 \to I_1$)

As the intermediate moment approaches the largest one, the growth rate (5) behaves as

$$\lim_{I_2 \to I_1} \sigma = \lim_{I_2 \to I_1} \Omega \sqrt{\frac{(I_1 - I_2)(I_2 - I_3)}{I_1 I_3}} = 0. \tag{7}$$

The body becomes a *symmetric top*, the ellipsoid gains a circular cross-section, and the unstable flip disappears (replaced by steady gyroscopic precession). This limit is not merely academic: Bubbar & Zhu measured $I_1 \approx I_2 \approx 78\ \mathrm{g\,m^2}$ for their racquet (with $I_3 \approx 1.39\ \mathrm{g\,m^2}$), yet still observed clean flips because the two moments were "sufficiently different" to keep σ nonzero. For a visually rapid flip in our figures we therefore use a well-separated set $I = (3, 2, 1)$ rather than the near-degenerate measured values.

3.2 High-Speed Limit and the Role of Air Drag

In the high-speed limit of Eq. (6), $\lim_{\omega_0 \to \infty} T = c$, a constant offset. Bubbar & Zhu note that the physical meaning of b and c is unclear (they imply periodic motion at $\omega_0 = 0$ and a finite period as $\omega_0 \to \infty$), suggesting the empirical model should not be extrapolated to extreme regimes.

Physical Conceptualization — the "torque-free" idealization.
Our entire derivation assumes *zero* external torque. In the real experiment of [1], the spinning racquet experiences a small aerodynamic drag torque because its cross-section is asymmetric; the authors measured a mean net torque of $-0.048 \pm 0.026\ \mathrm{N\,m}$, causing $|\mathbf{L}|$ to decay slowly. Gravity, by contrast, acts at the centre of mass and contributes *no* torque. Because the drag effect is small over a few flips, the conservative model still predicts the observed periods to within about 20%—a good example of judging an approximation by its measured consequences rather than discarding it outright.

4 Computational Verification

To confirm the analysis we wrote a self-contained Python simulation (NumPy / Matplotlib / SciPy). The companion interactive web version (Three.js) lets the reader vary I_1, I_2, I_3 and the initial conditions in real time.

4.1 The Numerical Integrator (RK4 + Quaternion)

Equation (2) has no simple closed form, so we integrate it numerically. We advance the seven-component state $(\boldsymbol{\omega}, q)$ with a fixed-step **fourth-order Runge–Kutta** scheme, whose local truncation error is $O(\Delta t^4)$, far smaller than the naive Euler method. In parallel we integrate the orientation quaternion via $\dot{q} = \frac{1}{2}\, q \otimes (0, \boldsymbol{\omega})$, renormalising $q \leftarrow q/|q|$ each step to prevent drift; quaternions avoid the gimbal-lock singularities of Euler angles. The loop is:

1. **Initialize** the inertia I, angular velocity $\boldsymbol{\omega}_0$, and quaternion q_0.
2. **RK4 step** for $\boldsymbol{\omega}_{t+\Delta t}$ from Eq. (2).
3. **Update orientation** $q_{t+\Delta t}$ from $\dot{q} = \frac{1}{2} q \otimes (0, \boldsymbol{\omega})$.
4. **Normalize** $q \leftarrow q/|q|$.

5. **Monitor** the invariants E and $|\mathbf{L}|$ as an accuracy check.

Running the integrator on a well-separated top gives relative drifts in E and $|\mathbf{L}|$ on the order of 10^{-13} over the full run—near the double-precision floor—confirming that the flips below are *physical*, not numerical artifacts.

4.2 Three-Axis Stability Comparison

Figure 1 starts the body spinning about each principal axis in turn with the same small perturbation. Only the intermediate axis is unstable: ω_2 periodically reverses sign while ω_1 and ω_3 surge, exactly as predicted by Eq. (5). Rotation about $\mathbf{r}_1$ and $\mathbf{r}_3$ stays flat.

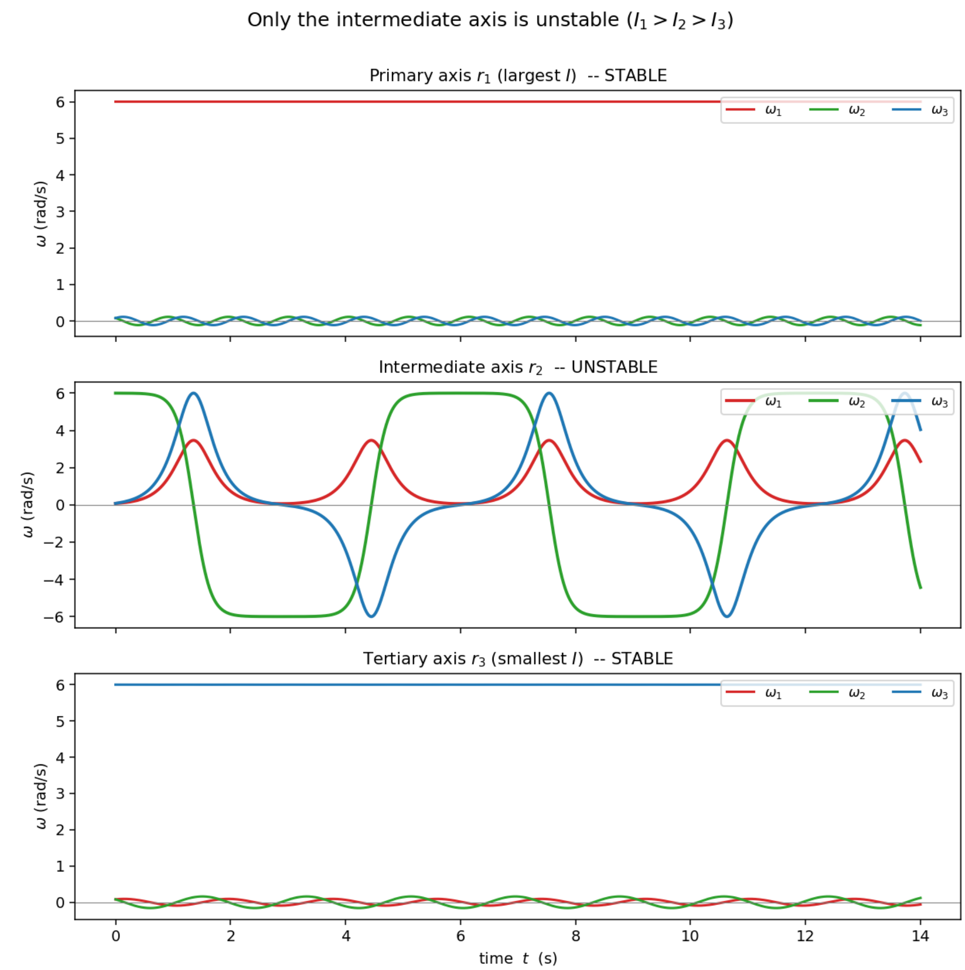

Figure 1: Angular-velocity components for rotation started about each of the three principal axes ($I = (3, 2, 1)$). Only the intermediate axis $\mathbf{r}_2$ exhibits the unstable flip.

4.3 The Flip Time Series

Figure 2 zooms in on the intermediate-axis case. The $\omega_2(t)$ curve resembles a square wave, switching sign at each flip, while ω_1 and ω_3 spike during the rapid 180° transitions—the periodic exchange of angular velocity that constitutes the tennis racquet effect.

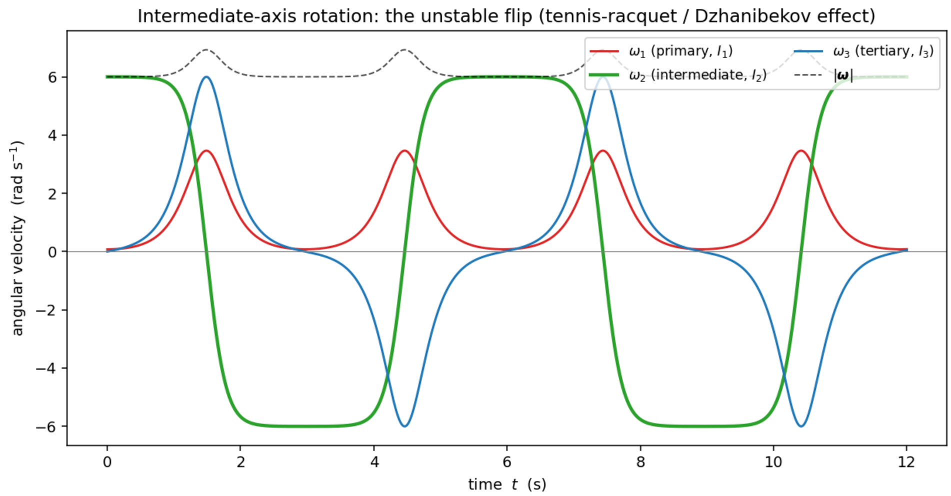

Figure 2: Time series of $\boldsymbol{\omega}(t)$ for intermediate-axis rotation: ω_2 flips sign periodically while ω_1, ω_3 spike.

4.4 The Sphere–Ellipsoid (Polhode) Geometry

Figure 3 renders the conservation picture of Eqs. (3)–(4): the motion lives on the intersection of the $|\mathbf{L}|$ sphere and the energy ellipsoid. About the major and minor axes the intersection is a small closed loop near a pole (stable); about the intermediate axis it is a *separatrix* that wraps the entire sphere—the geometric source of the instability.

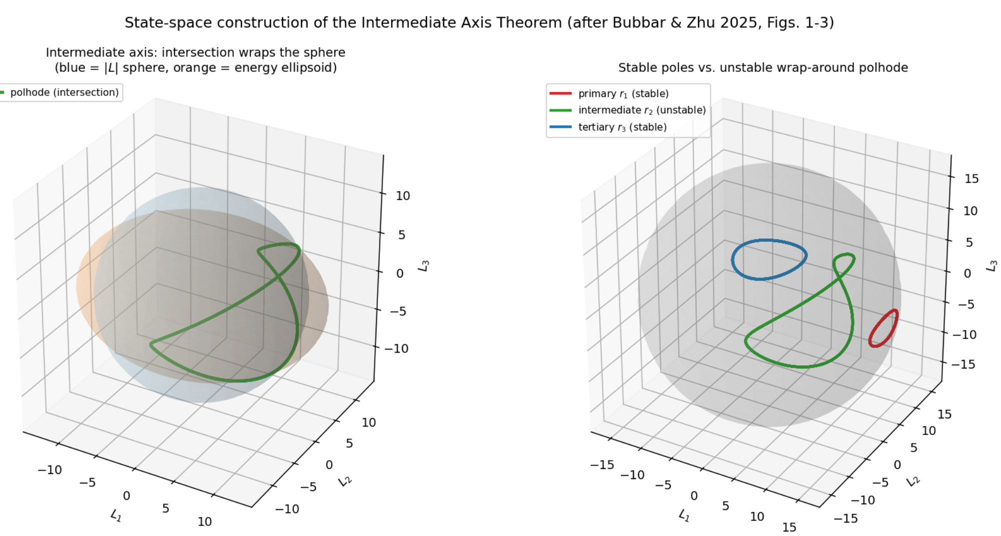

Figure 3: Left: the $|\mathbf{L}|$ sphere (blue) and energy ellipsoid (orange) with the intermediate-axis polhode wrapping the sphere. Right: stable polar loops versus the unstable wrap-around curve.

4.5 Flip Period Versus Initial Speed

Finally we test the model of Eq. (6). Measuring the flip period from the zero-crossings of ω_2 over a range of initial speeds and fitting the inverse model reproduces the predicted $T \propto 1/\omega_0$ trend

(Figure 4). The fitted parameters from our *synthetic* data differ from the racquet values of [1] ($a = 11.85,\ b = 2.30\ \text{s}^{-1},\ c = 0.047$ s, $\chi^2 = 1.58$ on 63 trials) because the simulated system uses the idealized $I = (3, 2, 1)$ top rather than the measured racquet, but the same inverse functional form holds.

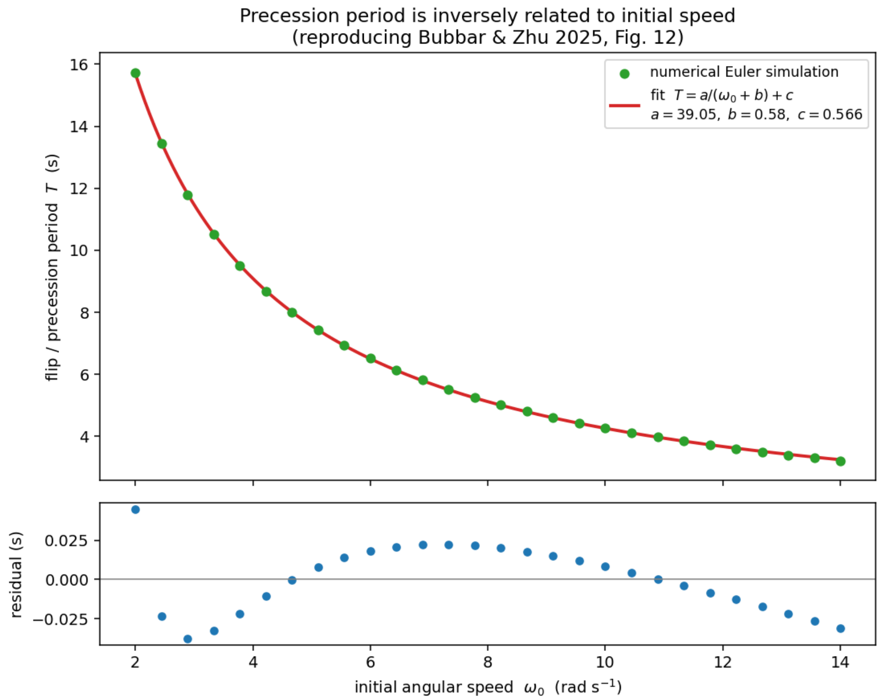

Figure 4: Flip period T versus initial angular speed ω_0 with the least-squares fit of $T = a/(\omega_0 + b) + c$; residuals shown below.

5 Critical Analysis and Commentary

- **Implicit assumptions.** The conservative model neglects air drag, which [1] showed is measurable (a slow $|\mathbf{L}|$ decay). It also assumes a perfectly rigid body and exactly diagonal inertia tensor; real objects flex and have product-of-inertia terms that we ignore.

- **Degeneracy near $I_1 \approx I_2$.** The measured racquet sits close to the symmetric-top limit, where our growth rate (5) is small; the threshold between "asymmetric" and "effectively symmetric" behaviour is subtle and was left open in [1].

- **Pedagogical effectiveness.** The geometric sphere-ellipsoid argument is visually compelling but hides the algebra; the linearisation in Section 2 is less elegant but makes the sign change unmistakable. Presenting both, as we do here, is more convincing than either alone.

- **Extensions and applications.** The same instability governs the tumbling of asteroids and the attitude control of spacecraft, where an unintended intermediate-axis spin has wrecked real satellites [7]. A natural extension is to add a small damping torque and watch the body migrate toward rotation about its axis of *maximum* inertia (the minimum-energy state at fixed $|\mathbf{L}|$).

6 Conclusion

We unpacked the intermediate axis theorem from first principles. Beginning with the torque-free Euler equations (2), we derived the energy and angular-momentum invariants and performed an explicit linear stability analysis showing that the characteristic coefficient changes sign only for the intermediate axis, yielding the exponential growth rate (5). A custom RK4-plus-quaternion simulation then confirmed every prediction: it reproduced the periodic 180° flip, conserved E and $|\mathbf{L}|$ to $\sim 10^{-13}$, displayed the separatrix polhode geometry, and recovered the inverse period–speed law of Bubbar & Zhu. Working through the intermediate mathematical steps—rather than accepting "it can be shown"—turned a baffling party trick into a transparent consequence of nonlinear coupling in rigid-body dynamics.

References

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